Fairuz Mubashwera

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Research Interests

Trustworthy and privacy-preserving machine learning, with a focus on **security of Vertical Federated Learning (VFL)** (label inference and gradient inversion attacks, and defenses against them); **machine unlearning in federated settings**, including standardized metrics and benchmarks for verifying unlearning; and **safety, reliability, and simulation-based testing of autonomous vehicles**, particularly the realism and diversity of pedestrian behavior in simulation.

Education

Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*
B.Sc. in Computer Science and Engineering — CGPA: 3.94/4.00 Jan. 2022 – Jun. 2026

- **Undergraduate Thesis:** *Security Vulnerabilities in Vertical Federated Learning: Label Inference Attacks and Defenses.*
- Consistently high term-wise performance across all eight terms (GPA range: 3.86–3.99); e.g., 3.99 (3rd Yr. T2), 3.98 (2nd Yr. T2), 3.96 (1st Yr. T2, 4th Yr. T2).

Research Experience

Label Inference Attacks in Vertical Federated Learning *BUET, Dhaka*
Undergraduate Thesis Research (Supervisor: [Supervisor Name]) 2025 – Present

- Investigating security vulnerabilities in Vertical Federated Learning (VFL), with a focus on **label inference** and **gradient inversion** attacks launched by semi-honest participants.
- Developed **SBEA**, an attack that exploits set bias in entity alignment to perform label inference and model extraction in VFL; manuscript submitted to *IJCAI 2026* (under review).
- Designing and empirically evaluating defense mechanisms that improve label privacy while preserving model utility in VFL pipelines.

Machine Unlearning in Vertical Federated Learning *BUET, Dhaka*
Independent Research 2026 – Present

- Studying **machine unlearning** in the VFL setting, where the vertical partitioning of features across parties makes existing (horizontal) unlearning approaches and evaluations inapplicable.
- Establishing a **standardized evaluation metric and benchmark protocol** for unlearning in VFL, quantifying unlearning completeness, residual information leakage, and post-unlearning model utility.
- Benchmarking representative unlearning strategies (retraining, gradient-based, and approximate methods) under the proposed metric to enable fair, reproducible comparison across VFL architectures.

Simulation-Based Testing of Autonomous Vehicles *BUET · Cal Poly · UC Santa Cruz*
Voluntary Research Collaboration 2024 – Present

- Working on enhancing the reliability of simulation-based testing frameworks for autonomous vehicles (AVs) in a joint collaboration between BUET, California Polytechnic State University (Cal Poly), and the University of California, Santa Cruz (UCSC).
- Developing **metrics to assess the diversity and realism of pedestrian trajectories** used in AV simulation scenarios, so that test suites better reflect real-world pedestrian behavior.

- Designed and deployed a public **trajectory annotation survey website** to collect human judgments of trajectory realism for metric validation.

Publications & Manuscripts

[1] **F. Mubashwera** et al., "SBEA: Exploiting Set Bias in Entity Alignment for Label Inference and Model Extraction in Vertical Federated Learning," manuscript submitted to *IJCAI 2026* (under review).

[2] **F. Mubashwera** et al., "Understanding Gendered Stereotype in Large Language Models: A Mini Review on Biasing and Debiasing Techniques," manuscript submitted to *Frontiers in Artificial Intelligence* (under review).

Professional & Teaching Experience

Presidency University — *Dhaka, Bangladesh*
Lecturer, Department of Computer Science and Engineering — Jun. 2026 – Present

- Teaching undergraduate courses in Computer Science and Engineering ([Course names, e.g., Programming, Data Structures]).
- Mentoring undergraduate students on coursework and research projects.

EliteResearchLab LLC — *Remote*
AI Research Intern — Jan. 2026 – Jun. 2026

- Worked on fairness auditing and bias-mitigation techniques for machine learning models, improving model transparency and interpretability.

Selected Projects

- **Trajectory Annotation Platform** — Web-based survey platform for crowdsourced annotation of pedestrian trajectory realism, supporting the AV simulation-testing research collaboration. *(ReactJS, Flask, PostgreSQL)*
- **AI Cheque Transaction & Fraud Detection System** — AI-driven cheque processing and fraud-detection pipeline designed for low-tier banks in Bangladesh; Finalist & Honourable Mention, Solvio AI Hackathon 2025. *(Python, PyTorch, OCR)*
- **MotherGPT** — Maternal-health web application providing AI-assisted health monitoring and personalized care guidance; Finalist, Dhaka Divisional Hackathon (IIEC, IUBAT) 2024. *(NLP, Flask, MongoDB)*
- **Urban Congestion Mobility Solution for Dhaka** — Proposed an innovative transportation solution for Dhaka's congestion problem; Finalist (Top 6), Honda International Symposium on Transportation and Vehicle 2025.

Technical Skills

Programming Languages	C, C++, Java, Python, SQL
ML / Research	PyTorch, Federated Learning frameworks, experiment design, simulation-based testing
Frameworks & Tools	SpringBoot, ReactJS, NodeJS, Flask, Docker, MongoDB, PostgreSQL, Git, LaTeX
Soft Skills	Academic mentorship, public speaking, leadership, policy advocacy, cross-cultural communication

Honors & Awards

- **Honda Young Engineer & Scientist's (Y-E-S) Award**, Honda Foundation, Japan, 2024 — prestigious award recognizing outstanding undergraduate students in science and engineering with exceptional innovation, leadership, and potential for societal impact.
- **Finalist (Top 6)**, Honda International Symposium on Transportation and Vehicle, 2025 — presented an innovative transportation solution for Dhaka's congestion problem.
- **Champion (Poster Presentation)**, BUET CSE Fest, 2024 — recognized for a research-based poster on technology.
- **Finalist & Honourable Mention**, Solvio AI Hackathon, 2025 — AI-driven cheque transaction and fraud-detection system.
- **Finalist**, Dhaka Divisional Hackathon, IIEC, IUBAT, 2024 — maternal-health AI application "MotherGPT."
- **Finalist**, Hult Prize at BUET, 2023 — sustainable apparel supply chain with technology.
- **Debating:** Breaking Team, United Asian Debating Championship (UADC) 2023; Champion, AUSTDC Nationals 2022; Runners-up, BUTEXDC Nationals 2022, BAUDS Nationals 2024, and KUET Open 2024.

Leadership & Academic Service

- **General Secretary**, BUET Debating Club (May 2025 – Present) — organized national-scale debate tournaments; led workshops on policy advocacy and public speaking; trained debaters at school, college, and university levels.
- **Institutional Representative**, United Asian Debating Union (Jan. 2025 – May 2025) — represented BUET in regional dialogue on the debate policy of the United Asian Debating Championship.
- **Executive Member**, Bangladesh Women in CSE (BWCSE) (Jan. 2023 – Present) — promoted women's participation and leadership in CSE through events, mentorship, and outreach; coordinated awareness sessions on sustainable technology practices.

References

[Supervisor Name]
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Bangladesh University of Engineering
and Technology (BUET), Dhaka
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(Thesis Supervisor)

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